These are not the only projects that are keeping many an ISRO scientist awake at night. Work is going on feverishly in its various laboratories across the country to realise ISRO's most ambitious mission yet: putting three Indian astronauts into orbit by indigenously building a 'human-rated' rocket launcher and crew module to fly them into space and bring them safely back to Earth. Called Gaganyaan, the mission requires ISRO to achieve mastery in a dozen new space disciplines, including mimicking Earth-like conditions in its crew module for the astronauts to stay on for several days and conduct experiments in zero gravity. The space agency is also testing a crew escape module that can eject the astronauts to safety in the event of a catastrophic failure of the launch vehicle that could endanger the lives of the astronauts at the launchpad during lift-off.

Meanwhile, the three astronauts for the mission have already been selected. All Indian Air Force pilots, they are undergoing the rigorous training required for being rocketed into space at 10x the speed of sound and then living in zero gravity conditions. Only three nations—the US, Russia and China—have sent up their own crewed spacecraft so far. Says ISRO chairman S. Somanath, "Many of the systems are new and we face a lot of technical challenges in realising them. We have to do multiple tests because the confidence-building process is very important. We will first test out all systems using unmanned flights, including sending a robot, before we launch our own astronauts into space. It has to be failure-proof because lives are involved."

In all these missions, ISRO is focusing on doing what it does best—technological innovation in a cost-effective manner. Aditya is a good example of how ISRO achieves this. To research the Sun, the agency will launch the spacecraft to an orbit 1.5 million km from Earth, or three times the distance from the Moon. Yet, with the Sun 150 million km away, Aditya will be only one per cent closer to it than Earth is. The plan is to place Aditya in the halo orbit around the Lagrange Point 1 or L1 where the gravitational pull of the Sun and Earth equals the centripetal force required for a spacecraft to move in tandem. This allows the spacecraft to observe the Sun continuously without consuming too much fuel. To get to the chosen Lagrange Point, ISRO is modifying its workhorse launcher PSLV to give it extra boost and put Aditya on track to get there. Further, as M. Sankaran, director, UR Rao Satellite Centre (URSC), explains, the mission team is using what are known as gravitational superhighways in space to propel Aditya to its destination the same way as the Chandrayaan launchers used Earth's gravity to slingshot it to the lunar orbit, thereby conserving fuel. This entails complex navigation and guidance, but ISRO has the experience of having

INTERVIEW

"INDIA IS NO LONGER THE LESSER COUSIN"

DR JITENDRA SINGH, the *Union Minister of State for Science & Technology and the Departments of Space and Atomic Energy*, was present at Mission Control when the Chandrayaan-3 spacecraft made a successful moon landing on August 23. *Group Editorial Director* RAJ CHENGAPPA spoke to him on the importance of the lunar mission and the future of India's space programme. Excerpts:

Q. What would you say is the importance of Chandrayaan to India's space programme and to the world?

Firstly, scientifically speaking, Chandrayaan-3 is going to get us inferences, conclusions and pictures, which would be possibly the first of their kind, for the entire scientific community across the world. Secondly, it has very evidently established India as a frontline nation as far as space technology is concerned. I must commend Team ISRO for having worked day and night for this mission, doing rehearsals and trials. They were working as if preparing to appear for a competitive exam...and that is the reason they were so confident at the end of the day. The rest of the world was not, but they were.

Q. You were with the scientists at Mission Control during the moon landing. What were your thoughts?

We were very conscious of the fact that the